

Representation and Storage of Digital Information

Laboratórios de Sistemas e Serviços

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Exercises

Exercise 1: Identifying Data Structure

For each of the following data sources, classify them as **Structured**, **Semi-structured**, or **Unstructured**:

1. A relational database table containing student grades.
 2. A collection of MP3 files from a podcast.
 3. A `config.yaml` file for a web server.
 4. A scanned PDF of a handwritten letter.
 5. A JSON response from a weather API.
 6. An Excel spreadsheet containing a list of products.
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Exercise 2: Character Encoding

1. How many bytes does the string "Hello!" take in ASCII?
 2. How many bytes does the string "Hello!" take in UTF-8?
 3. Why is UTF-8 preferred over ASCII for modern applications?
 4. What happens if you try to save the character "ç" in a file encoded as pure ASCII?
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Exercise 3: Processing Unstructured Data (Logs)

In this exercise, you will use Unix tools to extract information from a system log. Assume you have a file named `access.log` with the following content:

```
192.168.1.1 - - [20/Apr/2026:10:00:01] "GET /index.html" 200
192.168.1.5 - - [20/Apr/2026:10:00:05] "GET /images/logo.png" 200
192.168.1.1 - - [20/Apr/2026:10:00:10] "GET /contact.html" 200
192.168.1.10 - - [20/Apr/2026:10:00:15] "GET /admin" 403
192.168.1.1 - - [20/Apr/2026:10:00:20] "GET /index.html" 200
192.168.1.5 - - [20/Apr/2026:10:00:25] "GET /about.html" 200
```

1. Use `grep` to find all entries with a 403 status code.
 2. Use `awk` to print only the IP addresses (the first column).
 3. Combine `awk`, `sort`, and `uniq -c` to count how many requests each IP address made.
 4. Use `sed` to replace all occurrences of GET with POST in the output.
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Exercise 4: JSON, XML, and YAML

1. Convert the following XML snippet into a valid JSON object:

```
<sensor id="DHT11">
  <location>Room 101</location>
  <readings>
    <reading timestamp="2026-05-01T12:00:00">25.5</reading>
  </readings>
</sensor>
```

2. Represent the same information in YAML format.
 3. Use jq syntax to extract the location from your JSON object.
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Exercise 5: CSV Parsing

You have a CSV file `students.csv` with the following content:

```
id,name,grade,city
1,Alice,18,Aveiro
2,"Bob, Smith",14,Porto
3,Charlie,16,Aveiro
```

1. In row 2, why is the name field enclosed in double quotes?
 2. Use awk (with `-F ,`) to print only the names and grades.
 3. Calculate the average grade using awk.
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Exercise 6: Python and Pydantic

1. Write a Python script that reads a JSON file containing a list of users and validates them using a Pydantic model with `id` (int), `name` (str), and `email` (EmailStr).
2. What happens if one of the emails in the JSON is invalid?